

ALyl Chem 17 EQ P1 22w to 09s Paper 1 Carbonyl compounds 68marks

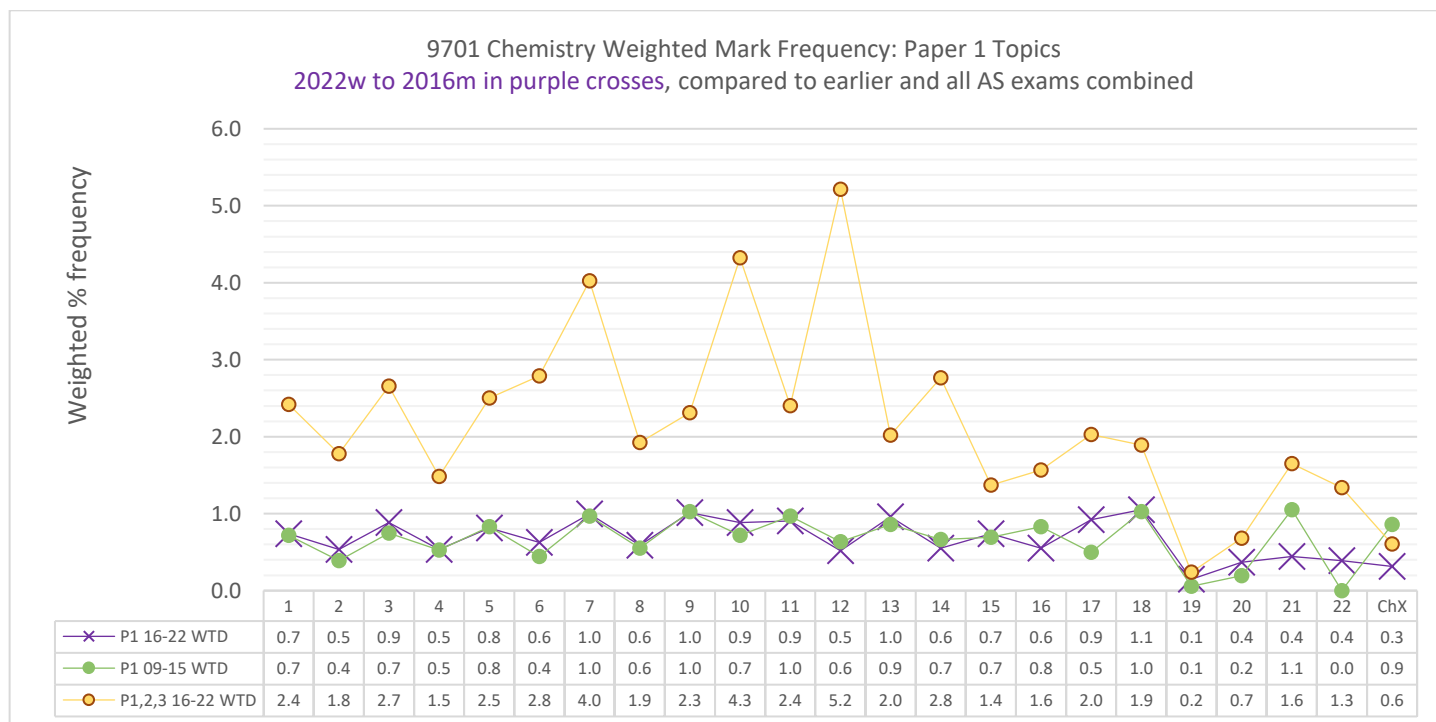
As you start and work through this worksheet you can tick off your progress to show yourself how much you have done, and what you need to do next. The first task is just to read the first question and should take you less than one minutes to complete.

Paper 1 Topic 17

Checklist Tick each task off as you go along

RANK:

P1 Noob	P1 Novice	P1 Bronze	P1 Silver	P1 Gold	P1 ¹ Winner	P1 Hero	P1 Legend
1 Q started	1 Q done	10% of marks	25% of marks	40% of marks	50% of marks	75% of marks	100% of marks
	1	7	17	27	34	51	68
	1	9	21	34	43	64	85



What the most thoughtful students will get out of their extensive studying will be a capacity to do meaningful brain-based work even under stressful conditions, which is a part of the self-mastery skillset that will continue to deliver value for the whole of their lives. Outstanding grades will also happen, but the most important goal from skillful action in study is being better at any important task, even if circumstances do not feel ideal.

As you are moving through your studies you can learn more about yourself by trying out new ways to manage yourself, and analysing how effective those new techniques were. In this reflective process not only will you get better at working positively and productively to deliver ambitious and successful outcomes, but you will be working towards one aspect of life's highest pursuit, summarised and inscribed on the Temple of Apollo at Delphi: "know thyself".

1. To complete these questions, as important as your answer, is checking your answer against the mark scheme.
2. For each page or group of 10 questions, convert your mark score into a percentage. This will allow you to see (and feel) your progress as you get more experience and understanding with each topic.
3. Multiple choice questions, done carefully where you explain and show yourself your thinking using written notes as you move through each question, can be more useful than just Paper 2 for students aiming for a C or B grade. Paper 2 should be the larger focus for students aiming for A and A* grades, however.
4. If you find you get a higher percentage answering short answer questions than multiple choice questions that often means you are NOT using the marking scheme correctly; your correct answer might not be fully complete for all the marks you are awarding. The marks easiest to miss rely on providing the largest amount of detail.

¹ **DO NOT** work on these higher levels of completion in your A2 year unless you have also achieved at least a "Silver" (25%) in the same topic in **Paper 2**, which is **MOST** of your **AS grade**, and Paper 3 which is a smaller part of your year but still important.

17 Carbonyl compounds

17.1 Aldehydes and ketones

Learning outcomes

Candidates should be able to:

- recall the reactions (reagents and conditions) by which aldehydes and ketones can be produced:
 - the oxidation of primary alcohols using acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or acidified KMnO_4 and distillation to produce aldehydes
 - the oxidation of secondary alcohols using acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or acidified KMnO_4 and distillation to produce ketones
- describe:
 - the reduction of aldehydes and ketones, using NaBH_4 or LiAlH_4 to produce alcohols
 - the reaction of aldehydes and ketones with HCN , KCN as catalyst, and heat to produce hydroxynitriles exemplified by ethanal and propanone
- describe the mechanism of the nucleophilic addition reactions of hydrogen cyanide with aldehydes and ketones in 17.1.2(b)
- describe the use of 2,4-dinitrophenylhydrazine (2,4-DNPH reagent) to detect the presence of carbonyl compounds
- deduce the nature (aldehyde or ketone) of an unknown carbonyl compound from the results of simple tests (Fehling's and Tollens' reagents; ease of oxidation)
- deduce the presence of a $\text{CH}_3\text{CO}-$ group in an aldehyde or ketone, $\text{CH}_3\text{CO}-\text{R}$, from its reaction with alkaline $\text{I}_2(\text{aq})$ to form a yellow precipitate of tri-iodomethane and an ion, RCO_2^-

Q# 1060/ AS Chemistry/2022/w/TZ 1/Paper 1/Q# 33//www.SmashingScience.org :o)

33 W reacts with alkaline $\text{I}_2(\text{aq})$ to form a yellow precipitate and $\text{CH}_3\text{CH}_2\text{CO}_2^-$ ions.

Which row identifies W and the yellow precipitate?

	identity of W	identity of yellow precipitate
A	butanone	CHI_3
B	butanone	CH_3I
C	propanone	CHI_3
D	propanone	CH_3I

Q# 1061/ AS Chemistry/2022/s/TZ 1/Paper 1/Q# 36//www.SmashingScience.org :o)

36 A carbonyl compound, X, reacts with HCN in the presence of NaCN to make a compound with M_r 85. Compound X does **not** react with Fehling's reagent.

What is compound X?

- butanal
- butanone
- propanal
- propanone



35 The skeletal formulae of two organic compounds are shown.



Which reagents can be used to distinguish these two compounds?

- 1 alkaline $I_2(aq)$
- 2 acidified $K_2Cr_2O_7$
- 3 2,4-dinitrophenylhydrazine (2,4-DNPH reagent)

A 1, 2 and 3 B 1 and 3 only C 2 and 3 only D 2 only

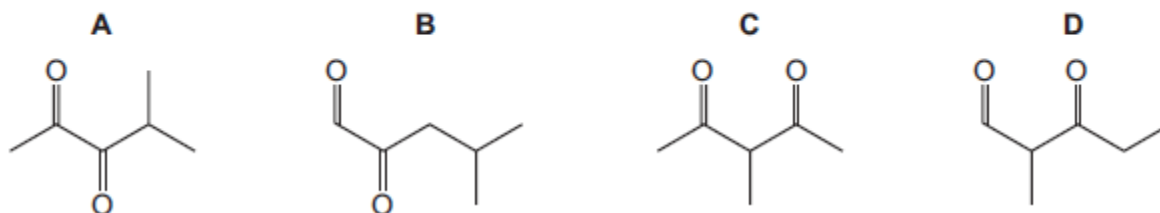
33 Which compound gives both:

- an orange precipitate with 2,4-DNPH reagent
- and a yellow precipitate with alkaline $I_2(aq)$?

- A ethanol
B methanal
C propanal
D propanone

27 Reduction of compound R with $LiAlH_4$ gives the compound 4-methylpentane-2,3-diol.

What could be the identity of compound R?



25 Alcohol Y gives a yellow precipitate with alkaline aqueous iodine. It can be oxidised to give a mixture of products including substance Z. Substance Z gives a red-brown precipitate with Fehling's solution.

Which alcohol could be Y?

- A $CH_3CH(OH)CH(CH_3)CH_2OH$
B $CH_3C(OH)(CH_3)CH_2CH_2OH$
C $CH_3CH(OH)CH_2CH(OH)CH_3$
D $CH_2(OH)CH_2CH(OH)CH_2CH_3$

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

39 Which reactions have a coloured organic product?

- 1 ethanal + 2,4-dinitrophenylhydrazine reagent
- 2 ethanol + acidified potassium dichromate(VI)
- 3 ethene + cold dilute acidified potassium manganate(VII)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

38 Which pairs of compounds may be distinguished by testing with alkaline aqueous iodine?

- 1 butanal and butanone
- 2 pentan-2-one and pentan-3-ol
- 3 propanone and propan-2-ol

28 Which statement is correct for the reaction of carbonyl compounds with HCN?

- A The reaction is catalysed by concentrated H_2SO_4 .
- B Pentan-2-one and HCN react to give a chiral product.
- C The reaction is a condensation reaction.
- D The reaction is nucleophilic substitution.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct



39 Two carbonyl compounds have the molecular formula C_3H_6O .

Which reagents give **different** observations with these two compounds?

- 1 acidified aqueous potassium manganate(VII)
- 2 Fehling's reagent
- 3 alkaline aqueous iodine

Q# 1070/ AS Chemistry/2021/m/TZ 2/Paper 1/Q# 28//www.SmashingScience.org :o)

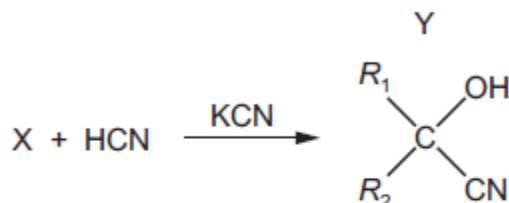
28 Ethanedioic acid has the formula HO_2CCO_2H .

What is the formula of aluminium ethanedioate?

- A AlC_2O_4 B $Al(C_2O_4)_3$ C $Al_2C_2O_4$ D $Al_2(C_2O_4)_3$

Q# 1071/ AS Chemistry/2021/m/TZ 2/Paper 1/Q# 26//www.SmashingScience.org :o)

26 The diagram shows the formation of compound Y from compound X in a chemical reaction. R_1 and R_2 are alkyl groups.

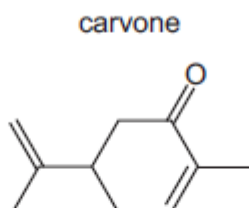


Which row about this reaction is correct?

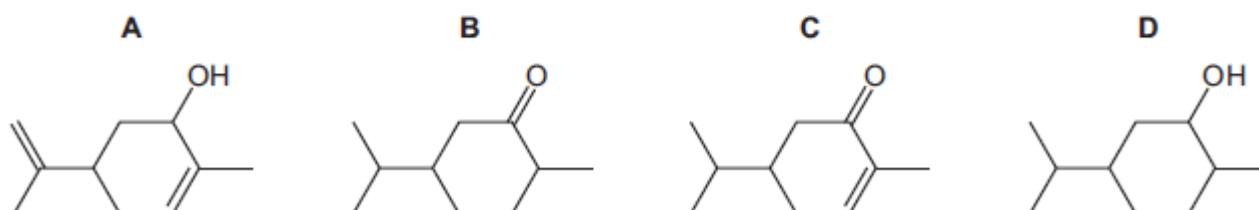
	mechanism	compound X
A	electrophilic addition	aldehyde
B	electrophilic addition	ketone
C	nucleophilic addition	ketone
D	nucleophilic addition	aldehyde

Q# 1072/ AS Chemistry/2020/w/TZ 1/Paper 1/Q# 28//www.SmashingScience.org :o)

28 Carvone is found in spearmint oil.



Which product is formed when carvone is reacted with $NaBH_4$?



27 Which pair of test results would prove that a substance, X, is a ketone?

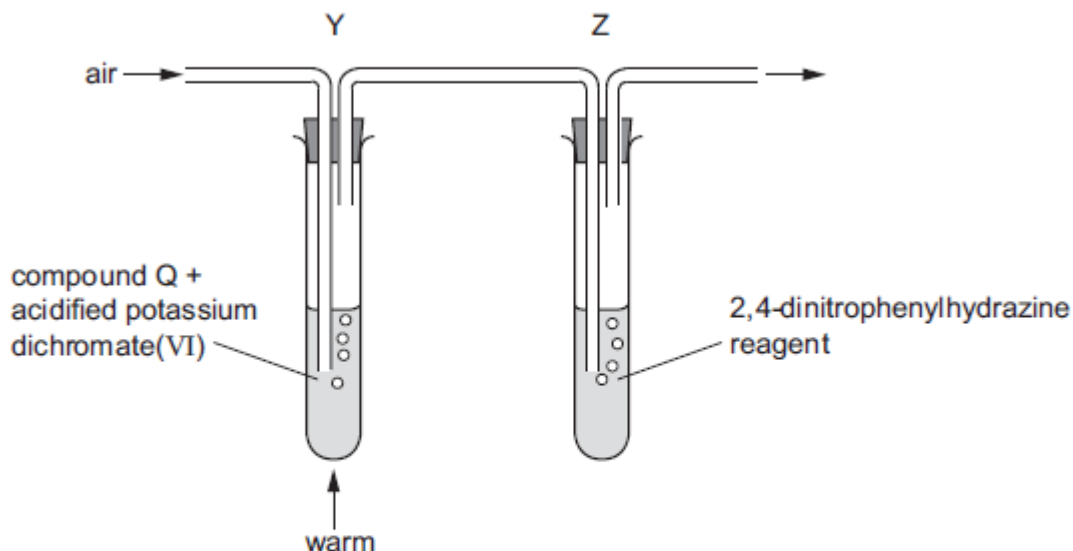
- A X has no reaction with Tollens' reagent. X reacts with alkaline aqueous iodine.
- B X is reduced by lithium aluminium hydride. X is oxidised by acidified dichromate(VI).
- C X reacts with 2,4-DNPH reagent. X has no reaction with Fehling's reagent.
- D X reacts with hydrogen cyanide. X is reduced by lithium aluminium hydride.

Q# 1074/ AS Chemistry/2020/s/TZ 1/Paper 1/Q# 38//www.SmashingScience.org :o)

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

38 When the apparatus is set up as shown, an orange precipitate forms in test-tube Z.



What could compound Q be?

- 1 $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- 2 $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
- 3 $(\text{CH}_3)_3\text{COH}$

Q# 1075/ AS Chemistry/2020/s/TZ 1/Paper 1/Q# 37//www.SmashingScience.org :o)

The responses A to D should be selected on the basis of

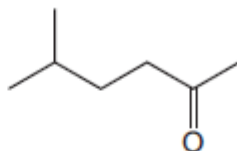
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

37 In which of the reactions is the organic compound oxidised by the given reagent?

- 1 $\text{CH}_3\text{CHO} + \text{HCN}$ reagent
- 2 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO} + \text{Tollens' reagent}$
- 3 $\text{CH}_3\text{CH}_2\text{CHO} + \text{Fehling's reagent}$

Q# 1076/ AS Chemistry/2020/s/TZ 1/Paper 1/Q# 20//www.SmashingScience.org :o)

20 The skeletal formula of compound X is shown.



Which row is correct?

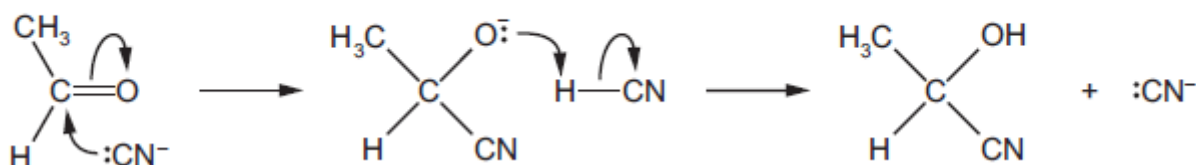
	molecular formula of X	observation on addition of X to Fehling's reagent
A	$\text{C}_7\text{H}_{14}\text{O}$	no change
B	$\text{C}_7\text{H}_{14}\text{O}$	red precipitate forms
C	$\text{C}_7\text{H}_{16}\text{O}$	no change
D	$\text{C}_7\text{H}_{16}\text{O}$	red precipitate forms

Q# 1077/ AS Chemistry/2020/m/TZ 2/Paper 1/Q# 37//www.SmashingScience.org :o)

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

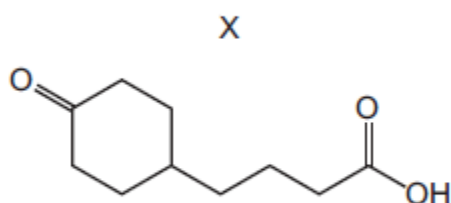
37 Ethanal and hydrogen cyanide react together. The reaction mechanism involves cyanide ions.



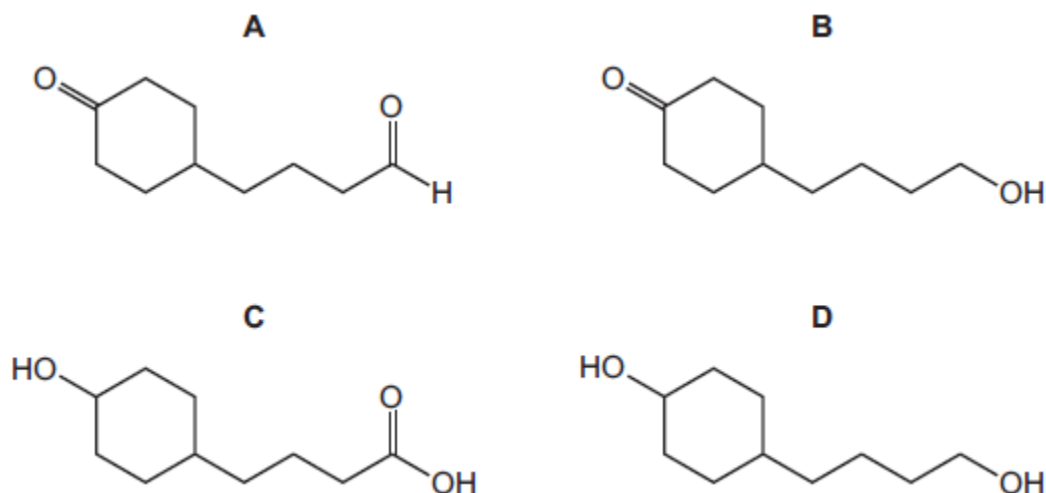
Which statements about this mechanism are correct?

- 1 CN^- acts as a catalyst.
- 2 CN^- is a nucleophile.
- 3 It is an addition reaction.

28 Compound X is treated with an excess of lithium aluminium hydride. The reaction is allowed to go to completion.



What is the structure of the organic product?



The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

39 Hydrogen cyanide reacts with propanone in the presence of potassium cyanide.

Which statements about this reaction are correct?

- 1 The cyanide ion is a catalyst for the reaction.
- 2 This is an addition reaction.
- 3 The intermediate behaves as a base.

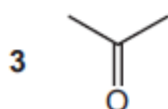
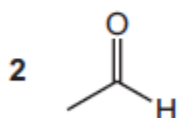
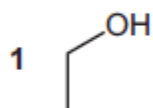
The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

38 Compound Y

- reacts with alkaline aqueous iodine to form a yellow precipitate
- changes the colour of warm, acidified potassium dichromate(VI) solution.

What could be compound Y?



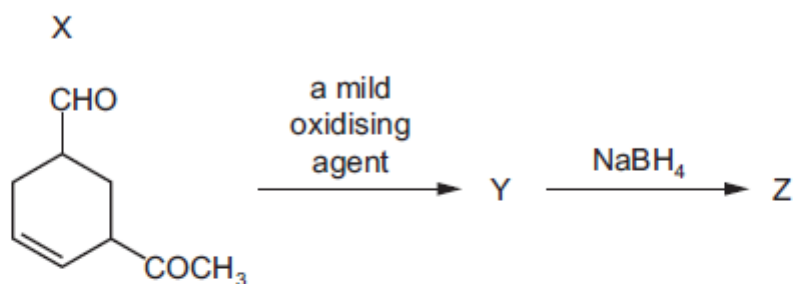
Q# 1081/ AS Chemistry/2019/s/TZ 1/Paper 1/Q# 29//www.SmashingScience.org :o)

29 Which reagent may be used to distinguish between propanone and ethanol?

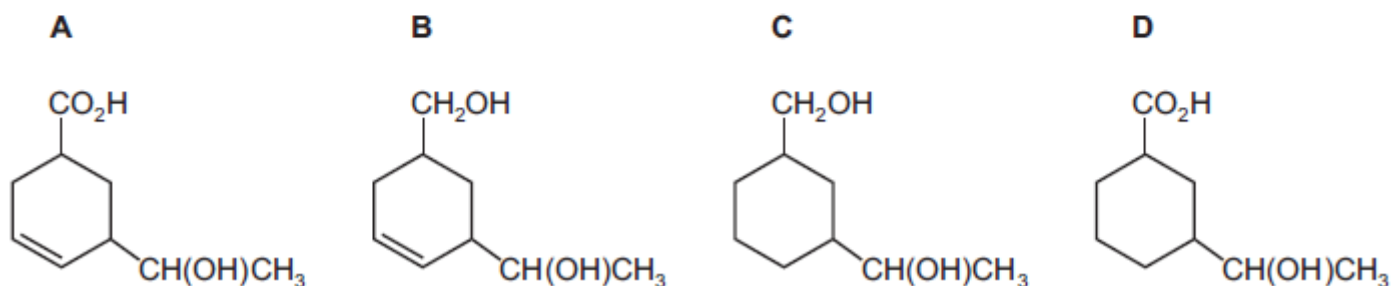
- A 2,4-dinitrophenylhydrazine
- B bromine water
- C Fehling's reagent
- D Tollens' reagent

Q# 1082/ AS Chemistry/2019/s/TZ 1/Paper 1/Q# 28//www.SmashingScience.org :o)

28 Compound X is treated with two reagents successively, forming compound Z.



What could be Z?



27 An organic compound T undergoes the following reactions.

- T is oxidised by hot, acidified potassium manganate(VII).
- T reacts with sodium to give hydrogen.

What could be compound T?

- A $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
 B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
 C $(\text{CH}_3)_3\text{COH}$
 D $\text{CH}_3\text{CH}_2\text{COCH}_3$

Q# 1084/ AS Chemistry/2019/m/TZ 2/Paper 1/Q# 39//www.SmashingScience.org :o)

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

39 The compounds listed are reacted with hydrogen cyanide.

Which compounds produce a molecule containing a chiral centre?

- 1 butanal
 2 butanone
 3 pentan-2-one

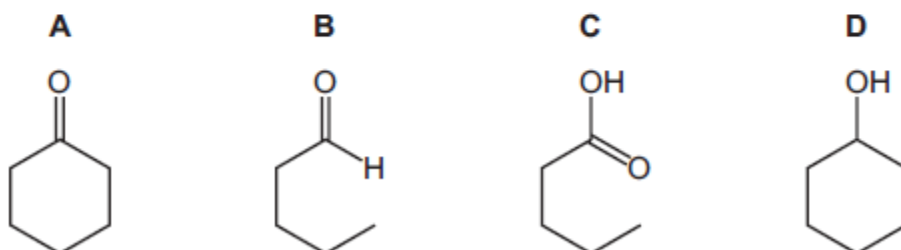
Q# 1085/ AS Chemistry/2019/m/TZ 2/Paper 1/Q# 28//www.SmashingScience.org :o)

28 When compound X is heated with $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$, a colour change from orange to green is observed.

Two tests are carried out on the organic product of this reaction.

test	result
Tollens' reagent	no change
2,4-dinitrophenylhydrazine	orange precipitate

What could be compound X?



27 Structural isomerism only should be considered when answering this question.

All the isomeric alcohols with the molecular formula $C_5H_{12}O$ are added separately to warm alkaline aqueous iodine.

How many of the isomers give a yellow precipitate?

- A 0 B 1 C 2 D 3

Q# 1087/ AS Chemistry/2018/w/TZ 1/Paper 1/Q# 40//www.SmashingScience.org :o)

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 Ethanal reacts with HCN in the presence of KCN.

Which changes in bonding occur during this reaction?

- 1 A carbon-carbon bond is formed.
- 2 A carbon-hydrogen bond is broken.
- 3 A carbon-nitrogen bond is broken.

Q# 1088/ AS Chemistry/2018/w/TZ 1/Paper 1/Q# 27//www.SmashingScience.org :o)

27 Which compound shows optical isomerism and gives a positive test with alkaline aqueous iodine?

- A $CH_3COCH(OH)CH_3$
 B $CH_3COCH_2CH_2OH$
 C $HOCH_2CH(CH_3)CHO$
 D $(CH_3)_2C(OH)CHO$

Q# 1089/ AS Chemistry/2018/s/TZ 1/Paper 1/Q# 40//www.SmashingScience.org :o)

The responses A to D should be selected on the basis of

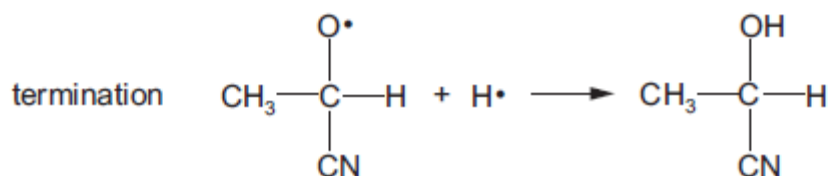
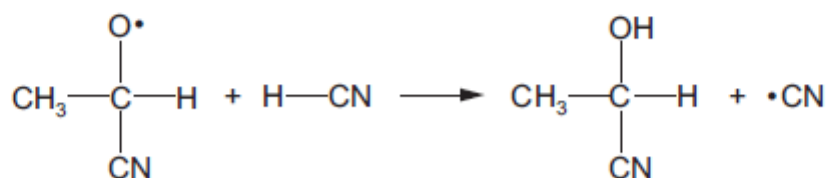
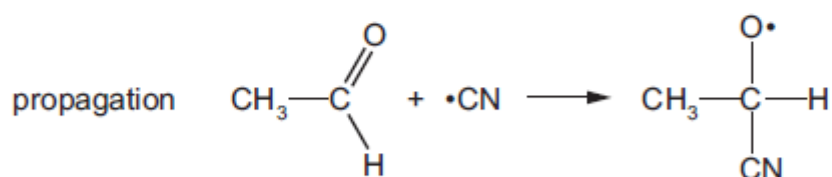
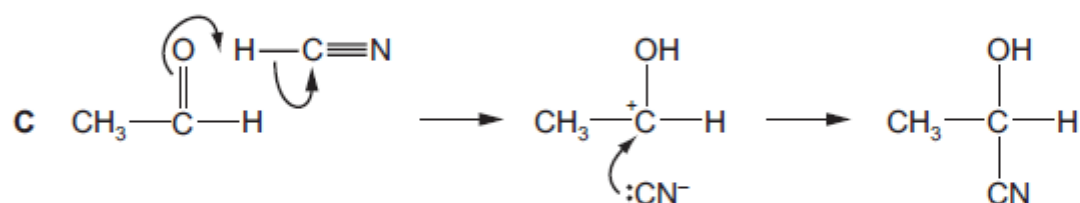
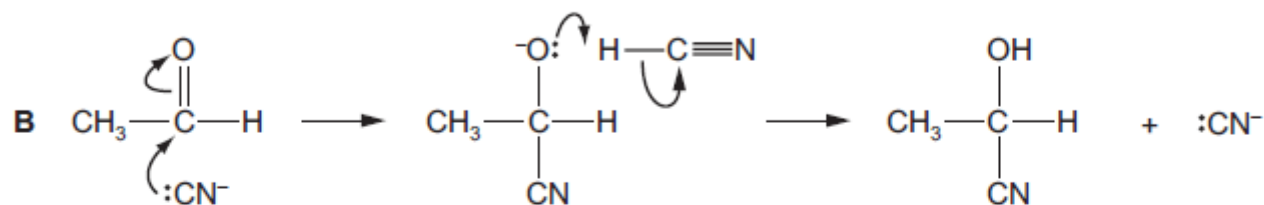
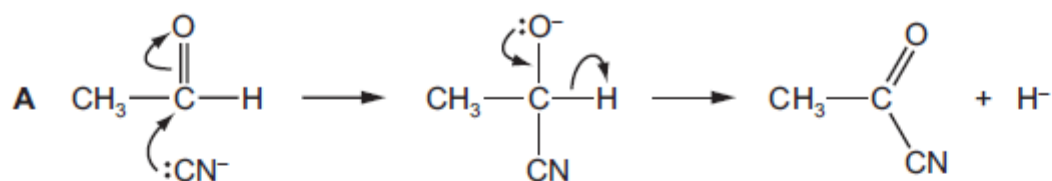
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 The M_r of compound X is 72. The composition by mass of X is 66.7% carbon, 11.1% hydrogen and 22.2% oxygen. X gives an orange precipitate with 2,4-dinitrophenylhydrazine reagent. X does not react with Fehling's reagent.

What can be deduced from this information?

- 1 X is a carbonyl compound.
- 2 X is a ketone.
- 3 X is butanone.

26 What is the mechanism for the reaction of ethanal, CH_3CHO , with hydrogen cyanide, HCN , in the presence of NaCN ?



25 Compound Q

- contains a chiral centre,
- gives a positive result with Fehling's reagent,
- gives a positive result with alkaline aqueous iodine.

What could compound Q be?

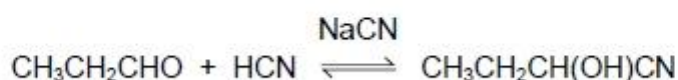
- A 1-hydroxybutanone
 B 2-hydroxybutanal
 C 3-hydroxybutanal
 D 3-hydroxybutanone

Q# 1092/ AS Chemistry/2018/m/TZ 2/Paper 1/Q# 39//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

- 39 Propanal reacts with hydrogen cyanide to form 2-hydroxybutanenitrile. A suitable catalyst for this reaction is sodium cyanide.



Which statements about the reaction of propanal with hydrogen cyanide are correct?

- HCN is a weaker nucleophile than the nucleophile provided by NaCN.
- The reaction mechanism involves two steps.
- The product of the reaction has a chiral carbon atom.

Q# 1093/ AS Chemistry/2018/m/TZ 2/Paper 1/Q# 26//www.SmashingScience.org :o)

- 26 Compound X produces a carboxylic acid when heated under reflux with acidified potassium dichromate(VI). Compound X does not react with sodium metal.

What could be the identity of compound X?

- A propanal
 B propanone
 C propan-1-ol
 D propan-2-ol



23 Which compound reacts with 2,4-dinitrophenylhydrazine reagent but does **not** react with Tollen's reagent?

- A $\text{CH}_3\text{COCO}_2\text{H}$
- B $\text{CH}_3\text{CH}(\text{OH})\text{CHO}$
- C CH_3COCHO
- D $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$

Q# 1095/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 40//

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 The reaction of ethanal, CH_3CHO , with HCN to form 2-hydroxypropanenitrile is catalysed by NaCN .

What are features of the intermediate of this reaction?

- 1 It is chiral.
- 2 It has a single negative charge on one of its atoms.
- 3 It is a nucleophile.

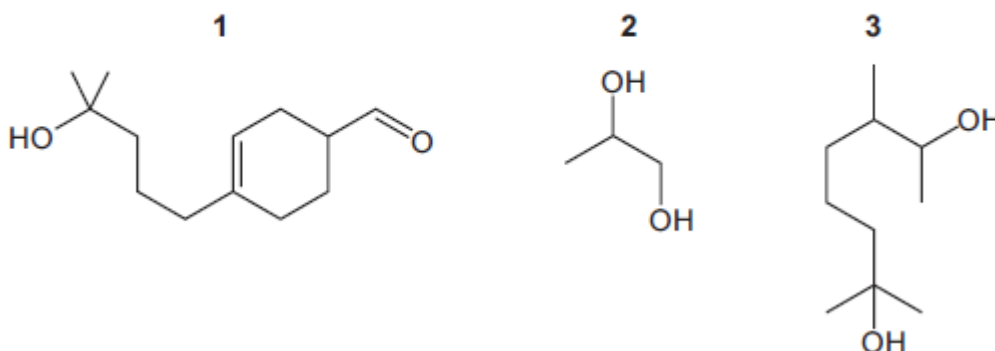
Q# 1096/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 39//

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

39 The compounds below are used to make perfumes.

Which compounds will produce a yellow precipitate with alkaline aqueous iodine?



Q# 1097/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 27//

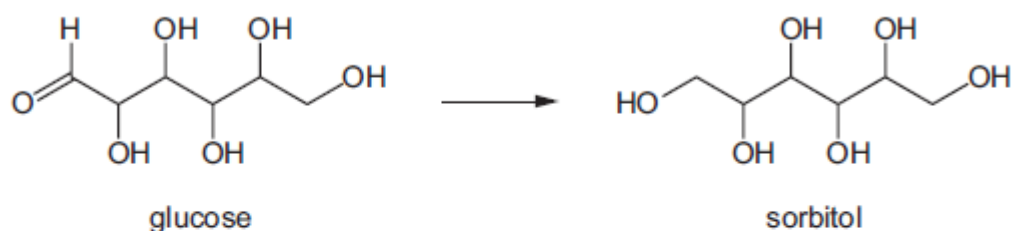
27 3-methylbutanone is treated with alkaline aqueous iodine. The mixture of products is then acidified.

Which compound is present in the final mixture of products?

- A 3-methylbutanoic acid
- B butanoic acid
- C methylpropanoic acid
- D propanoic acid

Q# 1098/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 26//

26 Glucose can be used to prepare sorbitol, a compound used as a sugar substitute.



Which reagent may be used for this conversion?

- A acidified potassium dichromate(VI)
- B sodium borohydride
- C sodium hydroxide
- D Tollens' reagent

Q# 1099/ AS Chemistry/2017/s/TZ 1/Paper 1/Q# 29//www.SmashingScience.org :o)

29 Which compound gives a positive test with alkaline aqueous iodine and does **not** show optical isomerism?

- A $\text{CH}_3\text{COCH}_2\text{CH}_2\text{OH}$
- B $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CHO}$
- C $\text{CH}_3\text{COCH}(\text{OH})\text{CH}_3$
- D $(\text{CH}_3)_2\text{C}(\text{OH})\text{CHO}$

Q# 1100/ AS Chemistry/2017/s/TZ 1/Paper 1/Q# 25//www.SmashingScience.org :o)

25 Diols in which both hydroxy groups are bonded to the same carbon atom spontaneously eliminate a molecule of water to produce a carbonyl compound.

Which compound is hydrolysed to form a product that gives a positive reaction with 2,4-dinitrophenylhydrazine but **not** with Fehling's reagent?

- A 1,1-dibromopropane
- B 1,2-dibromopropane
- C 1,3-dibromopropane
- D 2,2-dibromopropane



20 Structural isomerism and stereoisomerism should be considered when answering this question.

Each of the following carbonyl compounds is reacted with NaBH_4 . The product of each reaction is heated with Al_2O_3 at 600°C , generating one product or a mixture of isomers.

Which carbonyl compound will produce the most isomers?

- A butanal
- B butanone
- C pentan-3-one
- D propanone

Q# 1102/ AS Chemistry/2017/m/TZ 2/Paper 1/Q# 39//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

39 Which compounds react with alkaline aqueous iodine to give a pale yellow precipitate of tri-iodomethane?

- 1 butanone
- 2 ethanal
- 3 propan-2-ol

Q# 1103/ AS Chemistry/2017/m/TZ 2/Paper 1/Q# 29//www.SmashingScience.org :o)

29 Compound X, $\text{CH}_3\text{CH}(\text{OH})\text{CH}(\text{CHO})\text{CH}_3$, is heated under reflux with an excess of acidified $\text{K}_2\text{Cr}_2\text{O}_7$ to form compound Y.

Both X and Y are separately warmed with Fehling's solution and the observations noted.

What are the observations?

- A Both X and Y give a red precipitate.
- B Only X gives a red precipitate.
- C Only Y gives a red precipitate.
- D Neither X nor Y gives a red precipitate.

Q# 1104/ AS Chemistry/2017/m/TZ 2/Paper 1/Q# 28//www.SmashingScience.org :o)

28 $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$ reacts with hydrogen cyanide to form an organic product called a cyanohydrin.

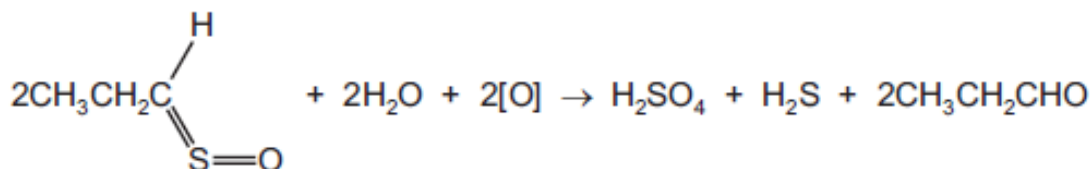
Which statement is correct?

- A The cyanohydrin product has one chiral centre.
- B The cyanohydrin product is formed by electrophilic addition.
- C The cyanohydrin product is formed via an intermediate which contains a C–OH group.
- D The formation of the cyanohydrin product requires the use of cyanide ions as a catalyst.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 When onions are peeled in air, the reaction shown is thought to occur.

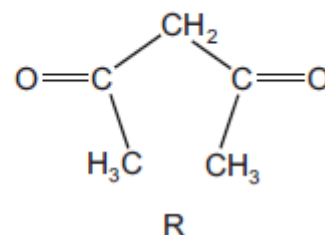
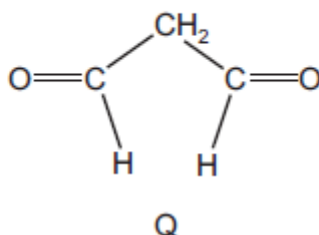
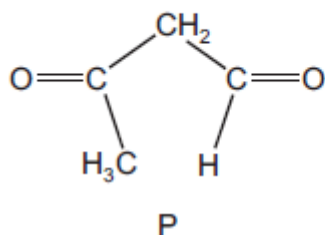


Which tests would give a positive reaction with the organic product?

- 1 warming with Tollens' reagent
- 2 warming with acidified potassium manganate(VII)
- 3 warming with alkaline aqueous iodine

Q# 1106/ AS Chemistry/2016/w/TZ 1/Paper 1/Q# 27//www.SmashingScience.org :o)

27 P, Q and R are carbonyl compounds.



Fehling's solution can be used to help identify these compounds.

Which compounds form a red-brown precipitate on warming with Fehling's solution?

- A** P, Q and R **B** P and Q only **C** P only **D** Q only

Q# 1107/ AS Chemistry/2016/s/TZ 1/Paper 1/Q# 28//www.SmashingScience.org :o)

28 Which reagent cannot be used to distinguish between ethanal and propanone?

- A acidified sodium dichromate(VI) solution
- B alkaline aqueous iodine
- C cold acidified potassium manganate(VII) solution
- D Fehling's reagent



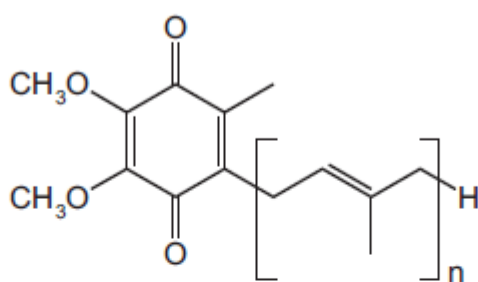
- 24 Alcohol Y gives product Z after mild oxidation. Z gives a positive result with Tollens' reagent and with 2,4-dinitrophenylhydrazine reagent.

What could be the identity of alcohol Y?

- A butan-1-ol
- B butan-2-ol
- C butan-2,3-diol
- D 2-methylbutan-2-ol

- 20 People who take statin drugs to control their blood cholesterol may also take 'coenzyme Q₁₀'.

The diagram shows a simplified structure of one form of this coenzyme.

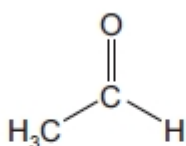


coenzyme Q₁₀

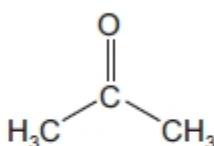
Which row describes this structure correctly?

	the coenzyme is	number of π bonds in one molecule
A	an aldehyde	$n + 2$
B	an aldehyde	$n + 4$
C	a ketone	$n + 2$
D	a ketone	$n + 4$

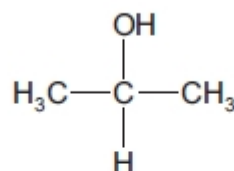
- 28 Tollens' reagent can be used to help identify compounds P, Q and R.



P



Q

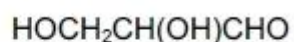


R

Which compound(s) form a silver precipitate on warming with Tollens' reagent?

- A P and Q
- B P only
- C Q only
- D R only

20 Which reagent will give a different observation with compounds P and Q?



P



Q

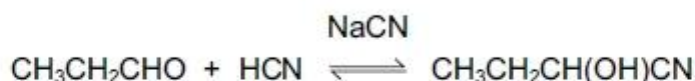
- A $\text{Br}_2(\text{aq})$
- B hot acidified KMnO_4
- C silver nitrate in ammonia solution
- D warm acidified $\text{K}_2\text{Cr}_2\text{O}_7$

Q# 1112/ AS Chemistry/2014/w/TZ 1/Paper 1/Q# 37//www.SmashingScience.org :o)

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

37 Propanal will react with hydrogen cyanide to form 2-hydroxybutanenitrile. A suitable catalyst for this reaction is sodium cyanide.



Which statements about the reaction of propanal with hydrogen cyanide are correct?

- 1 The CN^- ion attacks the propanal molecule to form an intermediate ion.
- 2 The product of the reaction has a chiral carbon atom.
- 3 The CN^- ion is a stronger electrophile than the HCN molecule.

Q# 1113/ AS Chemistry/2014/s/TZ 1/Paper 1/Q# 25//www.SmashingScience.org :o)

25 $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$ reacts with hydrogen cyanide to form an organic product called a cyanohydrin.

Which feature applies to the cyanohydrin product?

- A It has one chiral centre.
- B It is formed by electrophilic addition.
- C It is formed via an intermediate which contains the C–OH group.
- D Its formation requires the use of cyanide ions as a catalyst.

Q# 1114/ AS Chemistry/2013/s/TZ 1/Paper 1/Q# 38//www.SmashingScience.org :o)

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct



38 In which reactions is the organic compound oxidised by the given reagent?

- 1 $\text{CH}_3\text{CH}_2\text{CHO}$ + Fehling's reagent
- 2 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ + Tollens' reagent
- 3 CH_3CHO + 2,4-dinitrophenylhydrazine reagent

Q# 1115/ AS Chemistry/2012/w/TZ 1/Paper 1/Q# 28//www.SmashingScience.org :o)

28 In 1903 Arthur Lapworth became the first chemist to investigate a reaction mechanism. The reaction he investigated was that of hydrogen cyanide with propanone.

What do we now call the mechanism of this reaction?

- A electrophilic addition
- B electrophilic substitution
- C nucleophilic addition
- D nucleophilic substitution

Q# 1116/ AS Chemistry/2012/s/TZ 1/Paper 1/Q# 26//www.SmashingScience.org :o)



Which of these compounds can be oxidised by acidified dichromate(VI) solution and also gives a positive response to Tollens' reagent?

- A W and X only
- B W and Y only
- C X and Z only
- D Y and Z only

Q# 1117/ AS Chemistry/2012/s/TZ 1/Paper 1/Q# 25//www.SmashingScience.org :o)

25 Which compound, on reaction with hydrogen cyanide, produces a compound with a chiral centre?

- A CH_3CHO
- B $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$
- C $\text{CH}_3\text{CO}_2\text{CH}_3$
- D HCHO

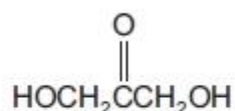
Q# 1118/ AS Chemistry/2011/w/TZ 1/Paper 1/Q# 38//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct



38 DHA is a colourless liquid which reacts with protein in skin to cause it to darken. It has the structure shown.



DHA

Which observations would be made when testing this substance?

- 1 Hydrogen is produced when sodium is added.
- 2 A coloured precipitate is produced when 2,4-dinitrophenylhydrazine reagent is added.
- 3 A silver precipitate is produced when Tollens' reagent is added.

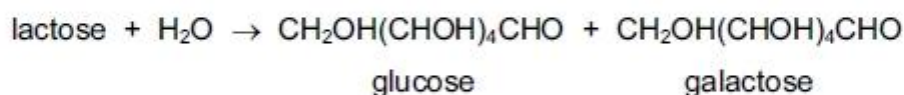
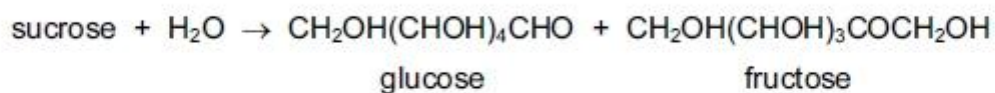
Q# 1119/ AS Chemistry/2011/w/TZ 1/Paper 1/Q# 37//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

37 Disaccharides, $C_{12}H_{22}O_{11}$, are important in the human diet. For example, sucrose is found in pastries and lactose occurs in milk products.

Both of these compounds can be hydrolysed.



Which statements about these hydrolysis products are correct?

- 1 Glucose and fructose have structural isomers.
- 2 Glucose and galactose are optical isomers.
- 3 Glucose and galactose are ketones.

Q# 1120/ AS Chemistry/2011/w/TZ 1/Paper 1/Q# 33//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct



33 The three statements that follow are all true.

Which of these can be explained, at least in part, by reference to hydrogen bonding?

- 1 At 0°C ice floats on water.
- 2 The boiling point of propan-2-ol is 82°C. The boiling point of propanone is 56°C.
- 3 At 20°C propanone and propanal mix completely.

Q# 1121/ AS Chemistry/2011/s/TZ 1/Paper 1/Q# 29//www.SmashingScience.org :o)

29 Aldehydes and ketones are carbonyl compounds.

Which of them react with NaBH_4 and react with Fehling's reagent?

- A both aldehydes and ketones
- B aldehydes only
- C ketones only
- D neither aldehydes nor ketones

Q# 1122/ AS Chemistry/2011/s/TZ 1/Paper 1/Q# 28//www.SmashingScience.org :o)

28 A compound Y is treated with warm acidified potassium dichromate(VI). The resulting organic product gives an orange precipitate with 2,4-dinitrophenylhydrazine reagent but does not give a silver mirror with Tollens' reagent.

What is Y?

- A butan-1-ol
- B butan-2-ol
- C butanal
- D 2-methylpropan-2-ol

Q# 1123/ AS Chemistry/2011/s/TZ 1/Paper 1/Q# 26//www.SmashingScience.org :o)

26 Compound X changes the colour of warm acidified sodium dichromate(VI) from orange to green. 1 mol of X reacts with 2 mol of HCN in the presence of KCN.

What could X be?

- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
- B $\text{CH}_3\text{COCH}_2\text{COCH}_3$
- C $\text{H}_2\text{C}=\text{CHCH}_2\text{CHO}$
- D $\text{OHCCH}_2\text{CH}_2\text{CHO}$

Q# 1124/ AS Chemistry/2010/s/TZ 1/Paper 1/Q# 24//www.SmashingScience.org :o)

24 What is formed when propanone is refluxed with a solution of NaBH_4 ?

- A propanal
- B propan-1-ol
- C propan-2-ol
- D propane



Q# 1125/ AS Chemistry/2010/s/TZ 1/Paper 1/Q# 23//www.SmashingScience.org :o)

23 Which reagent will give similar results with both butanone and butanal?

- A acidified aqueous potassium dichromate(VI)
- B an alkaline solution containing complexed Cu^{2+} ions (Fehling's solution)
- C an aqueous solution containing $[\text{Ag}(\text{NH}_3)_2]^+$ (Tollens' reagent)
- D 2,4-dinitrophenylhydrazine reagent

Q# 1126/ AS Chemistry/2009/w/TZ 1/Paper 1/Q# 40//www.SmashingScience.org :o)

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 A compound has a relative molecular mass of 88 and its molecule contains only four carbon atoms.

What could this compound be?

- 1 a saturated non-cyclic diol
- 2 a secondary alcohol containing an aldehyde group
- 3 a primary alcohol containing a ketone group

Q# 1127/ AS Chemistry/2009/s/TZ 1/Paper 1/Q# 26//www.SmashingScience.org :o)

26 Glycol, used in anti-freeze, has the formula $\text{HOCH}_2\text{CH}_2\text{OH}$. It can be oxidised to give a number of products.

What is the molecular formula of an oxidation product of glycol that will **not** react with sodium?

- A $\text{C}_2\text{H}_2\text{O}_2$ B $\text{C}_2\text{H}_2\text{O}_3$ C $\text{C}_2\text{H}_2\text{O}_4$ D $\text{C}_2\text{H}_4\text{O}_2$

Mark Scheme ALV1 Chem 17 EQ P1 22w to 09s Paper 1 Carbonyl compounds 68marks

Q# 1060/ AS Chemistry/2022/w/TZ 1/Paper 1/Q#

33//www.SmashingScience.org :o)

33 | A

Q# 1061/ AS Chemistry/2022/s/TZ 1/Paper 1/Q#

36//www.SmashingScience.org :o)

36 | D

Q# 1062/ AS Chemistry/2022/s/TZ 1/Paper 1/Q#

35//www.SmashingScience.org :o)

35 | B

Q# 1063/ AS Chemistry/2022/m/TZ 2/Paper 1/Q#

33//www.SmashingScience.org :o)

33 | D

Q# 1064/ AS Chemistry/2021/w/TZ 1/Paper 1/Q#

27//www.SmashingScience.org :o)

27 | A

Q# 1065/ AS Chemistry/2021/w/TZ 1/Paper 1/Q#

25//www.SmashingScience.org :o)

25 | A

Q# 1066/ AS Chemistry/2021/s/TZ 1/Paper 1/Q#

39//www.SmashingScience.org :o)

39 | D

Q# 1067/ AS Chemistry/2021/s/TZ 1/Paper 1/Q#

38//www.SmashingScience.org :o)

38 | B

Q# 1068/ AS Chemistry/2021/s/TZ 1/Paper 1/Q#

28//www.SmashingScience.org :o)

28 | B



Q# 1069/ AS Chemistry/2021/m/TZ 2/Paper 1/Q#

39//www.SmashingScience.org :o)

40 | B

Q# 1070/ AS Chemistry/2021/m/TZ 2/Paper 1/Q#

28//www.SmashingScience.org :o)

28 | D

Q# 1071/ AS Chemistry/2021/m/TZ 2/Paper 1/Q#

26//www.SmashingScience.org :o)

26 | C

Q# 1072/ AS Chemistry/2020/w/TZ 1/Paper 1/Q#

28//www.SmashingScience.org :o)

28 | A

Q# 1073/ AS Chemistry/2020/w/TZ 1/Paper 1/Q#

27//www.SmashingScience.org :o)

27 | C

Q# 1074/ AS Chemistry/2020/s/TZ 1/Paper 1/Q#

38//www.SmashingScience.org :o)

38 | B

Q# 1075/ AS Chemistry/2020/s/TZ 1/Paper 1/Q#

37//www.SmashingScience.org :o)

37 | C

Q# 1076/ AS Chemistry/2020/s/TZ 1/Paper 1/Q#

20//www.SmashingScience.org :o)

20 | A

Q# 1077/ AS Chemistry/2020/m/TZ 2/Paper 1/Q#

37//www.SmashingScience.org :o)

37 | A

Q# 1078/ AS Chemistry/2020/m/TZ 2/Paper 1/Q#

28//www.SmashingScience.org :o)

28 | D

Q# 1079/ AS Chemistry/2019/w/TZ 1/Paper 1/Q#

39//www.SmashingScience.org :o)

39 | A

Q# 1080/ AS Chemistry/2019/s/TZ 1/Paper 1/Q#

38//www.SmashingScience.org :o)

38 | B

Q# 1081/ AS Chemistry/2019/s/TZ 1/Paper 1/Q#

29//www.SmashingScience.org :o)

29 | A

Q# 1082/ AS Chemistry/2019/s/TZ 1/Paper 1/Q#

28//www.SmashingScience.org :o)

28 | A

Q# 1083/ AS Chemistry/2019/s/TZ 1/Paper 1/Q#

27//www.SmashingScience.org :o)

27 | A

Q# 1084/ AS Chemistry/2019/m/TZ 2/Paper 1/Q#

39//www.SmashingScience.org :o)

39 | A

Q# 1085/ AS Chemistry/2019/m/TZ 2/Paper 1/Q#

28//www.SmashingScience.org :o)

28 | D

Q# 1086/ AS Chemistry/2019/m/TZ 2/Paper 1/Q#

27//www.SmashingScience.org :o)

27 | C

Q# 1087/ AS Chemistry/2018/w/TZ 1/Paper 1/Q#

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40 | B

Q# 1088/ AS Chemistry/2018/w/TZ 1/Paper 1/Q#

27//www.SmashingScience.org :o)

27 | A

Q# 1089/ AS Chemistry/2018/s/TZ 1/Paper 1/Q#

40//www.SmashingScience.org :o)

40 | A

Q# 1090/ AS Chemistry/2018/s/TZ 1/Paper 1/Q#

26//www.SmashingScience.org :o)

26 | B

Q# 1091/ AS Chemistry/2018/s/TZ 1/Paper 1/Q#

25//www.SmashingScience.org :o)

25 | C

Q# 1092/ AS Chemistry/2018/m/TZ 2/Paper 1/Q#

39//www.SmashingScience.org :o)

39 | A

Q# 1093/ AS Chemistry/2018/m/TZ 2/Paper 1/Q#

26//www.SmashingScience.org :o)

26 | A

Q# 1094/ AS Chemistry/2018/m/TZ 2/Paper 1/Q#

23//www.SmashingScience.org :o)

23 | A

Q# 1095/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 40//

40 | A

Q# 1096/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 39//

39 | C

Q# 1097/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 27//

27 | C

Q# 1098/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 26//

26 | B

Q# 1099/ AS Chemistry/2017/s/TZ 1/Paper 1/Q#

29//www.SmashingScience.org :o)

29 | A

Q# 1100/ AS Chemistry/2017/s/TZ 1/Paper 1/Q#

25//www.SmashingScience.org :o)

25 | D

Q# 1101/ AS Chemistry/2017/s/TZ 1/Paper 1/Q#

20//www.SmashingScience.org :o)

20 | B



Q# 1102/ AS Chemistry/2017/m/TZ 2/Paper 1/Q#
39//www.SmashingScience.org :o)

39 | A

Q# 1103/ AS Chemistry/2017/m/TZ 2/Paper 1/Q#
29//www.SmashingScience.org :o)

29 | B

Q# 1104/ AS Chemistry/2017/m/TZ 2/Paper 1/Q#
28//www.SmashingScience.org :o)

28 | D

Q# 1105/ AS Chemistry/2016/w/TZ 1/Paper 1/Q#
40//www.SmashingScience.org :o)

40 B

Q# 1106/ AS Chemistry/2016/w/TZ 1/Paper 1/Q#
27//www.SmashingScience.org :o)

27 B

Q# 1107/ AS Chemistry/2016/s/TZ 1/Paper 1/Q#
28//www.SmashingScience.org :o)

28 B

Q# 1108/ AS Chemistry/2016/s/TZ 1/Paper 1/Q#
24//www.SmashingScience.org :o)

24 A

Q# 1109/ AS Chemistry/2016/m/TZ 2/Paper 1/Q#
20//www.SmashingScience.org :o)

20 D

Q# 1110/ AS Chemistry/2015/w/TZ 1/Paper 1/Q#
28//www.SmashingScience.org :o)

28 B

Q# 1111/ AS Chemistry/2015/s/TZ 1/Paper 1/Q#
20//www.SmashingScience.org :o)

20 C

Q# 1112/ AS Chemistry/2014/w/TZ 1/Paper 1/Q#
37//www.SmashingScience.org :o)

37 B

Q# 1113/ AS Chemistry/2014/s/TZ 1/Paper 1/Q#
25//www.SmashingScience.org :o)

25 D

Q# 1114/ AS Chemistry/2013/s/TZ 1/Paper 1/Q#
38//www.SmashingScience.org :o)

38 B

Q# 1115/ AS Chemistry/2012/w/TZ 1/Paper 1/Q#
28//www.SmashingScience.org :o)

28 C

Q# 1116/ AS Chemistry/2012/s/TZ 1/Paper 1/Q#
26//www.SmashingScience.org :o)

26 C

Q# 1117/ AS Chemistry/2012/s/TZ 1/Paper 1/Q#
25//www.SmashingScience.org :o)

25 A

Q# 1118/ AS Chemistry/2011/w/TZ 1/Paper 1/Q#
38//www.SmashingScience.org :o)

38 B

Q# 1119/ AS Chemistry/2011/w/TZ 1/Paper 1/Q#
37//www.SmashingScience.org :o)

37 B

Q# 1120/ AS Chemistry/2011/w/TZ 1/Paper 1/Q#
33//www.SmashingScience.org :o)

33 B

Q# 1121/ AS Chemistry/2011/s/TZ 1/Paper 1/Q#
29//www.SmashingScience.org :o)

29 B

Q# 1122/ AS Chemistry/2011/s/TZ 1/Paper 1/Q#
28//www.SmashingScience.org :o)

28 B

Q# 1123/ AS Chemistry/2011/s/TZ 1/Paper 1/Q#
26//www.SmashingScience.org :o)

26 D

Q# 1124/ AS Chemistry/2010/s/TZ 1/Paper 1/Q#
24//www.SmashingScience.org :o)

24 C

Q# 1125/ AS Chemistry/2010/s/TZ 1/Paper 1/Q#
23//www.SmashingScience.org :o)

23 D

Q# 1126/ AS Chemistry/2009/w/TZ 1/Paper 1/Q#
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40 C

Q# 1127/ AS Chemistry/2009/s/TZ 1/Paper 1/Q#
26//www.SmashingScience.org :o)

26 A

